

Introduction & Explanation

SmartERP - Billing, Inventory & Accounting Management System

(Tally-Inspired Web Application)

MVP FEATURES TO FOCUS ON :

LEDGER - SUPPLIER
CUSTOMER
ITEM / STOCK

VOUCHERS - SALES / CUSTOMER BILL
PURCHASE / INDIRECT STOCK ENTRY

1. Introduction

SmartERP is a cloud-based Billing, Inventory, Accounting, and Business Management System inspired by Tally but built using modern web technologies.

The application enables businesses to manage:

- Accounting
- Inventory
- Billing
- Purchases
- Sales
- GST
- Financial Reports
- Customer Management
- Supplier Management
- Banking Transactions

through a web browser while preserving Tally's keyboard-first workflow.

Unlike traditional ERP systems that heavily rely on mouse interactions, SmartERP is designed to be fully keyboard-driven, allowing accountants and operators to work efficiently using shortcut keys.

The application supports up to 5 companies per user account and provides role-based access control, real-time reporting, and modern cloud deployment capabilities.

2. Project Objective

The primary objective is to build a Tally-inspired web application that:

- Manages complete business accounting
 - Handles inventory management
 - Supports billing and invoicing
 - Maintains GST compliance
 - Generates financial reports
 - Operates efficiently through keyboard shortcuts
 - Supports multiple companies
-

3. Industry Relevance

ERP and accounting software are used by:

- Retail Stores
- Medical Shops
- Manufacturers
- Distributors
- Wholesalers
- Chartered Accountants
- Small and Medium Businesses

Popular software includes:

- TallyPrime
- Zoho Books

- Busy Accounting
- QuickBooks

Building this project demonstrates:

- Enterprise software development
- Complex business workflows
- Financial systems understanding
- Database design
- Full-stack engineering

This type of project is significantly more valuable on a resume than typical CRUD applications because it mirrors software used by real businesses.

Tech STack

4. Tech Stack

Frontend

- Next.js
- React.js
- Tailwind CSS
- ShadCN UI
- TanStack Table
- React Hook Form

Backend

- Node.js
- Express.js

Database

- PostgreSQL / Supabase

Authentication

- JWT Authentication / Supabase

PDF Generation

- PDFKit

Excel Export

- ExcelJS

Deployment

- Vercel (Frontend)
- Railway/Render (Backend)
- PostgreSQL Database

Application Flow

5. Application Flow

Step 1: Authentication

User logs in using:

- Email
- Password

OR

- Admin-created account

After successful login:

User is redirected to Company Selection Screen.

Step 2: Company Selection

Each account can manage a maximum of 5 companies.

User can:

- Create Company
- Select Existing Company
- Alter Company
- Delete Company

Company Information:

- Company Name
 - Address
 - GST Number
 - Financial Year
 - State
 - Contact Information
-

Step 3: Gateway of SmartERP

After selecting a company:

User enters the main dashboard.

Menu Structure:

Gateway of SmartERP

Masters

Transactions

Inventory

Accounting

Banking

Payroll

GST

Reports

Utilities

Administration

Master Module

6. Masters Module

Ledger Creation

Create:

- Customer Ledger
- Supplier Ledger
- Expense Ledger
- Income Ledger
- Bank Ledger
- Cash Ledger
- Stock Item Ledger

Features:

- Create Ledger
 - Alter Ledger
 - Delete Ledger
 - Search Ledger
-

Group Creation

Create Groups:

- Assets
 - Liabilities
 - Income
 - Expenses
-

Stock Groups

Examples:

Electronics

Furniture

Groceries

Medical

Stock Items

Store:

- Item Name
 - SKU
 - Purchase Price
 - Selling Price
 - Quantity
 - GST Percentage
-

Units of Measure

Examples:

- PCS
 - KG
 - BOX
 - LTR
-

7. Voucher Management

Contra Voucher

Cash ↔ Bank Transfers

Payment Voucher

Record payments.

Examples:

- Rent
- Electricity
- Salaries

Receipt Voucher

Record received money.

Journal Voucher

Accounting adjustments.

Purchase Voucher

Supplier purchases.

Inventory increases automatically.

Sales Voucher

Customer sales.

Inventory decreases automatically.

Invoice generated automatically.

Credit Note

Sales return management.

Debit Note

Purchase return management.

8. Inventory Management

Features:

- Stock In
- Stock Out
- Stock Transfer
- Inventory Adjustment
- Inventory Valuation

Inventory Tracking:

Current Stock

Reserved Stock

Damaged Stock

Available Stock

9. Billing System

Generate:

- GST Invoice
- Proforma Invoice
- Quotation
- Estimate

Features:

- Auto Invoice Number
 - PDF Download
 - Print Invoice
 - Email Invoice - Dont provide this feature
-

10. Customer Management

Store:

- Name
- Mobile Number
- Address
- Outstanding Balance

Features:

- Add Customer
 - Edit Customer
 - Customer Ledger
 - Customer Statements
-

11. Supplier Management

Store:

- Supplier Information
 - Purchase History
 - Payment History
 - Outstanding Dues
-

12. Banking Module

Features:

- Bank Transactions
 - Reconciliation
 - Fund Transfers
 - Cheque Management
-

13. GST Module

Features:

- GST Calculation
- GST Reports
- GSTR Summary

GST Types:

- CGST
 - SGST
 - IGST
-

14. Reports Module

Financial Reports

- Balance Sheet
 - Profit & Loss
 - Trial Balance
 - Cash Flow Statement
-

Inventory Reports

- Stock Summary
- Low Stock Report
- Item Movement Report

Sales Reports

- Daily Sales
- Monthly Sales
- Top Customers

Purchase Reports

- Purchase Register
- Supplier Summary

GST Reports

- Tax Summary
 - GST Register
-
-

Short Keys

15. Keyboard-Only Navigation System

The entire application should be usable without a mouse.

Global Shortcuts

F1 = Company Selection

F2 = Change Financial Year

F3 = Company Information

F4 = Calculator

F5 = Refresh

ESC = Previous Screen

CTRL + Q = Logout

CTRL + H = Home

CTRL + K = Command Search

Masters Shortcuts

ALT + L = Create Ledger

ALT + A = Alter Ledger

ALT + G = Create Group

ALT + S = Create Stock Item

ALT + U = Unit Creation

Voucher Shortcuts

F6 = Receipt Voucher

F7 = Journal Voucher

F8 = Sales Voucher

F9 = Purchase Voucher

F10 = Reversing Journal

ALT + F8 = Credit Note

ALT + F9 = Debit Note

Inventory Shortcuts

CTRL + I = Inventory Dashboard

CTRL + N = New Item

CTRL + E = Edit Item

CTRL + D = Delete Item

CTRL + T = Stock Transfer

CTRL + R = Stock Report

Billing Shortcuts

CTRL + B = New Invoice

CTRL + P = Print Invoice

CTRL + SHIFT + P = PDF Download

CTRL + M = Email Invoice

Customer Shortcuts

CTRL + C = New Customer

CTRL + SHIFT + C = Customer Ledger

Supplier Shortcuts

CTRL + S = New Supplier

CTRL + SHIFT + S = Supplier Ledger

Reports Shortcuts

ALT + B = Balance Sheet

ALT + P = Profit & Loss

ALT + T = Trial Balance

ALT + C = Cash Flow

ALT + R = Stock Summary

ALT + X = GST Report

Search & Navigation

CTRL + F = Search

CTRL + SHIFT + F = Global Search

ARROW KEYS = Navigation

ENTER = Select

TAB = Next Field

SHIFT + TAB = Previous Field

Database Modules

16. Database Modules

Users

Companies

Ledgers

Groups

Stock Groups

Stock Items

Units

Customers

Suppliers

Invoices

Voucher Entries

Transactions

Inventory Transactions

GST Records

Reports

Audit Logs

2 Week Schedule & Resources

17. Two-Week Development Plan

Week 1

Day 1

- Requirement Analysis
- Database Design

Day 2

- Backend Setup
- PostgreSQL Setup

Day 3

- Authentication Module

Day 4

- Company Management

Day 5

- Dashboard UI

Day 6

- Ledger Management

Day 7

- Group Management

Week 2

Day 8

- Stock Management

Day 9

- Purchase Voucher

Day 10

- Sales Voucher

Day 11

- Billing System

Day 12

- Reports Module

Day 13

- Keyboard Shortcut Integration

Day 14

- Testing & Deployment
-

18. Learning Resources

Frontend

Next.js Documentation

<https://nextjs.org/docs>

React Documentation

<https://react.dev>

Tailwind Documentation

<https://tailwindcss.com/docs>

Backend

Node.js Documentation

<https://nodejs.org/docs>

Express Documentation

<https://expressjs.com>

Database

PostgreSQL Documentation

<https://www.postgresql.org/docs>

YouTube Resources

Code with Antonio

JavaScript Mastery

freeCodeCamp

Traversy Media

Hitesh Choudhary

Akshay Saini

GitHub References

<https://github.com/topics/erp>

<https://github.com/topics/accounting-software>

<https://github.com/topics/inventory-management>

<https://github.com/topics/billing-system>

Tally Tutorial

 Tally Class - Lecture 1 | टैली कैसे सीखें | Tally ERP 9 Course in Hindi

Future Enhancements

19. Future Enhancements

- Barcode Scanner
 - Multi-Branch Management
 - Mobile Application
 - WhatsApp Invoice Sharing
 - Role-Based Permissions
 - Audit Logs
 - AI Business Insights
 - OCR Bill Scanning
 - Bank API Integration
 - UPI Reconciliation
-

20. Conclusion

SmartERP is a modern cloud-based ERP platform inspired by Tally that combines Accounting, Inventory Management, Billing, GST, Banking, Reporting, and Business Analytics into a single web application.

The project demonstrates advanced full-stack development skills, enterprise software architecture, financial workflow understanding, database design, reporting systems, and keyboard-first user experience design.

For a fresher or full-stack developer, this project is a high-impact portfolio project because it reflects the complexity of real-world business software used daily by thousands of organizations.

Jargons

1. Ledger

Meaning

A ledger is an account where all transactions related to a person, item, bank, expense, or income are recorded.

Example

Customer: Rahul Traders

01 Aug Sale ₹5,000

05 Aug Payment ₹2,000

Balance Due = ₹3,000

This complete record is Rahul Traders' ledger.

2. Customer Ledger

Meaning

Stores all transactions related to customers.

Example

Customer purchases products on credit.

Customer Name: ABC Store

Purchased:

₹10,000

Paid:

₹4,000

Outstanding:

₹6,000

3. Supplier Ledger

Meaning

Stores all transactions related to suppliers/vendors.

Example

Supplier: XYZ Wholesalers

Purchased Goods:

₹50,000

Paid:

₹30,000

Outstanding:

₹20,000

4. Stock Item

Meaning

A product that is bought and sold.

Example

Product:

Laptop

Current Quantity:

15

5. Stock Ledger

Meaning

Tracks movement of inventory.

Example

Opening Stock = 100

Purchased = 20

Sold = 15

Closing Stock = 105

Formula:

Opening Stock + Purchases - Sales = Closing Stock

6. Voucher

Meaning

A document representing a transaction.

Example

Sold Laptop

Amount:

₹50,000

A Sales Voucher gets created.

7. Sales Voucher

Meaning

Records product sales.

Effect

Stock ↓

Customer Due ↑

Revenue ↑

Example

Sold:

Laptop

Qty: 2

Price: ₹40,000

Sales Voucher created.

8. Purchase Voucher

Meaning

Records purchases from suppliers.

Effect

Stock ↑

Supplier Due ↑

Example

Purchased:

20 Keyboards

Stock increases.

9. Invoice

Meaning

A bill generated during sale.

Example

Invoice No: INV-1001

Customer:
ABC Store

Total:
₹12,000

10. Bill

Meaning

Proof of transaction.

Example

Customer receives bill after purchasing goods.

11. Outstanding Amount

Meaning

Money that has not been paid yet.

Example

Purchase:
₹10,000

Paid:
₹4,000

Outstanding:
₹6,000

12. Credit Sale

Meaning

Goods sold now, payment later.

Example

Sold:

₹20,000

Received:

₹0

Customer owes money.

13. Cash Sale

Meaning

Goods sold and payment received immediately.

Example

Sold:

₹20,000

Received:

₹20,000

14. Credit Purchase

Meaning

Goods purchased and payment made later.

Example

Bought:

₹50,000

Paid:

₹0

Supplier balance increases.

15. Quantity

Meaning

Number of units available.

Example

Keyboard = 50 pcs

16. Unit

Meaning

Measurement used for inventory.

Examples

PCS

KG

LTR

BOX

PACK

17. Opening Stock

Meaning

Stock available at the beginning.

Example

Laptop = 25

18. Closing Stock

Meaning

Stock available at the end.

Example

Opening = 50

Purchase = 20

Sales = 10

Closing = 60

19. Purchase Rate

Meaning

Price at which item is purchased.

Example

Keyboard

Purchase Price:

₹500

20. Selling Rate

Meaning

Price at which item is sold.

Example

Selling Price:
₹800

21. Profit

Meaning

Money earned after sale.

Formula

Profit
=
Selling Price
-
Purchase Price

Example

Purchase = ₹500

Sale = ₹800

Profit = ₹300

22. SKU (Stock Keeping Unit)

Meaning

Unique code for a product.

Example

LAP-001

MOU-001

KEY-001

23. Stock In

Meaning

Inventory increases.

Example

Purchase 10 laptops.

Stock In = +10

24. Stock Out

Meaning

Inventory decreases.

Example

Sell 2 laptops.

Stock Out = -2

25. Inventory

Meaning

All products available in the business.

Example

Laptop
Mouse
Keyboard
Monitor

26. Inventory Valuation

Meaning

Total value of inventory.

Formula

Quantity × Purchase Price

Example

100 Keyboards

Purchase Price:

₹500

Value:

₹50,000

27. GST

Meaning

Tax applied on goods/services.

Types

- CGST
- SGST
- IGST

Example

Product Price:

₹1000

GST:

18%

Total:

₹1180

28. HSN Code

Meaning

Government code used for GST classification.

Example

Laptop

HSN:

84713010

29. Party

Meaning

In accounting, "Party" refers to Customer or Supplier.

Example

ABC Traders

is a Party.

30. Closing Balance

Meaning

Final amount due.

Example

Customer Purchased:

₹10,000

Paid:
₹7,000

Closing Balance:
₹3,000
